

# Version 3 - Stacking Model from V2 Features - Feature Used / Dropped Audit

เทียบ source clean dataset กับ df\_engineered และระบุว่า feature ใดใช้ train model / feature ใดตัดทิ้ง

## Model Summary

| Model                          | Dataset           | df_engineered   | Used Features | Accuracy | Recall | F1     | AUC    | Rating |
|--------------------------------|-------------------|-----------------|---------------|----------|--------|--------|--------|--------|
| Stacking XGB + LGBM + CatBoost | df_engineered.csv | df_featured.csv | 38            | 61.60%   | 63.76% | 52.65% | 71.90% | B      |

## Source Artifacts

| artifact          | path                                                       | columns | sample_rows_read |
|-------------------|------------------------------------------------------------|---------|------------------|
| source_data       | docs\version 2\data\features\df_engineered.csv             | 61      | 5                |
| df_engineered     | docs\version 3\stacking_model_v3\data\df_featured.csv      | 40      | 5                |
| used_feature_list | docs\version 3\stacking_model_v3\data\v3_used_features.csv | 1       | 5                |

## Used Feature Groups

| group                 | feature_count | example_features                                                                                     |
|-----------------------|---------------|------------------------------------------------------------------------------------------------------|
| time                  | 11            | avg_delivery_days, customer_age_days, customer_tenure_months, days_since_last_order, delay_days      |
| other                 | 7             | courier_name, courier_type, is_repurchased_item, logistics_risk, promo_name                          |
| price_discount        | 7             | campaign_discount_pct, discount_applied_amount, promo_discount_rate, tier_discount_pct, total_amount |
| customer_profile      | 5             | age, age_group, damage_rate, gender, membership_tier                                                 |
| product               | 3             | brand, category, product_rating                                                                      |
| customer_history      | 2             | hist_order_count, hist_return_rate                                                                   |
| order_channel_payment | 2             | channel_type, payment_method                                                                         |
| geography             | 1             | province                                                                                             |

## Used Features (sample)

age, age\_group, avg\_delivery\_days, brand, campaign\_discount\_pct, category, channel\_type, courier\_name, courier\_type, customer\_age\_days, customer\_tenure\_months, damage\_rate, days\_since\_last\_order, delay\_days, delivery\_days, delivery\_time\_expected\_days, discount\_applied\_amount, gender, hist\_order\_count, hist\_return\_rate, is\_repurchased\_item, is\_weekend, logistics\_risk, membership\_tier, order\_dayofweek, order\_hour, order\_month, payment\_method, product\_rating, promo\_discount\_rate, promo\_name, promo\_type, province, quantity, tier\_discount\_pct, total\_amount, total\_discount\_pct, unit\_price

## Dropped / Not Used Feature Logic

- ตัด leakage/post-event/target fields จำนวนประมาณ 1 feature เช่น is\_returned
- ตัด identifier/high-cardinality fields จำนวนประมาณ 0 feature เช่น
- feature ที่เป็น raw date หรือ raw text จะถูกแปลงเป็น numeric/encoded/rolling feature ก่อน หรือไม่ใช้ถ้าไม่ช่วย performance
- feature ที่เพิ่มแล้ว performance ไม่ดีขึ้นหรือซ้ำซ้อนกับ feature สำคัญ จะถูกจัดเป็น NOT USED เพื่อลด resource/query cost

## Feature Audit Detail (sample rows)

| Feature                | In Clean | In Engineered | Used  | Status   | Reason                                                           |
|------------------------|----------|---------------|-------|----------|------------------------------------------------------------------|
| age                    | True     | True          | True  | USED     | model input feature                                              |
| age_group              | True     | True          | True  | USED     | model input feature                                              |
| avg_delivery_days      | True     | True          | True  | USED     | model input feature                                              |
| brand                  | True     | True          | True  | USED     | model input feature                                              |
| campaign_discount_pct  | True     | True          | True  | USED     | model input feature                                              |
| category               | True     | True          | True  | USED     | model input feature                                              |
| channel_type           | True     | True          | True  | USED     | model input feature                                              |
| courier_name           | True     | True          | True  | USED     | model input feature                                              |
| courier_type           | True     | True          | True  | USED     | model input feature                                              |
| customer_age_days      | True     | True          | True  | USED     | model input feature                                              |
| customer_tenure_months | True     | True          | True  | USED     | model input feature                                              |
| damage_rate            | True     | True          | True  | USED     | model input feature                                              |
| category_channel       | True     | False         | False | DROPPED  | raw field replaced by engineered/encoded feature or not required |
| category_payment       | True     | False         | False | DROPPED  | raw field replaced by engineered/encoded feature or not required |
| dataset_split          | False    | True          | False | NOT USED | engineered candidate not selected for this model                 |
| discount_amount_ratio  | True     | False         | False | DROPPED  | raw field replaced by engineered/encoded feature or not required |
| hist_order_count_180d  | True     | False         | False | DROPPED  | raw field replaced by engineered/encoded feature or not required |
| hist_order_count_30d   | True     | False         | False | DROPPED  | raw field replaced by engineered/encoded feature or not required |
| hist_order_count_365d  | True     | False         | False | DROPPED  | raw field replaced by engineered/encoded feature or not required |
| hist_order_count_60d   | True     | False         | False | DROPPED  | raw field replaced by engineered/encoded feature or not required |
| hist_order_count_90d   | True     | False         | False | DROPPED  | raw field replaced by engineered/encoded feature or not required |
| hist_return_count_180d | True     | False         | False | DROPPED  | raw field replaced by engineered/encoded feature or not required |
| hist_return_count_30d  | True     | False         | False | DROPPED  | raw field replaced by engineered/encoded feature or not required |
| hist_return_count_365d | True     | False         | False | DROPPED  | raw field replaced by engineered/encoded feature or not required |
| hist_return_count_60d  | True     | False         | False | DROPPED  | raw field replaced by engineered/encoded feature or not required |
| hist_return_count_90d  | True     | False         | False | DROPPED  | raw field replaced by engineered/encoded feature or not required |
| hist_return_rate_180d  | True     | False         | False | DROPPED  | raw field replaced by engineered/encoded feature or not required |
| hist_return_rate_30d   | True     | False         | False | DROPPED  | raw field replaced by engineered/encoded feature or not required |
| hist_return_rate_365d  | True     | False         | False | DROPPED  | raw field replaced by engineered/encoded feature or not required |
| hist_return_rate_60d   | True     | False         | False | DROPPED  | raw field replaced by engineered/encoded feature or not required |

Full CSV: docs\version 3\feature\_documentation\v3\_feature\_used\_unused\_audit.csv